



Sir Jagadish Chandra Bose at the Royal Institution, London

used by Marconi in his widely acclaimed demonstration of transatlantic communication, for which the invention of the radio is popularly credited to him in the year 1901. Whereas Sir Jagadish Chandra Bose had given a public demonstration of EM waves remotely ringing a bell and exploding some gunpowder. He actually holds the patent for the first solid-state diode to receive electromagnetic waves and created a number of now common microwave components.

During the foundation of the Bose Institute in 1917, he apparently made his dislike of patents quite evident. Over the years, there has been a lot of speculation regarding this being a case of simultaneous invention. While it is true that a lot of work was being done around wireless communication and EM waves around that time, Marconi has actually acknowledged the significance of Bose's work, albeit sparsely, in his writing.

Apart from being a physicist, Sir J.C. Bose was also a polymath, biologist, biophysicist, botanist and archaeologist. He made some significant contributions to the world of plant biology and even has a crater on the moon named after him.



EMAIL

Talking about controversies, the subject of the original inventor of email is actually quite the pickle, having even been debated on twitter. ARPANET was created with the objective of establishing text communication between two computers. Ray Tomlinson wrote a program that allowed this communication system to overcome the limitation of servers and send messages across servers using the @ sign that we are so familiar with now.



Shiva Ayyadurai

About seven years after this program, Shiva Ayyadurai, as a 14 year old in 1978 created a program that replicated the features of a traditional paper based mailing system in the form of currently familiar features like 'Inbox', 'Outbox', 'Drafts', 'Subject' etc. Ayyadurai also held the copyright to the term EMAIL at the time of this creation. A point of interest is that patents were not handed out for software at the time of this creation.

While there is still some room for discussion regarding whose contribution to email was greater, there is no doubt in the fact that email would have been vastly different had it not had Shiva's contributions. That being said, the contribution of Indians to email does not end here!

One of the world's first free webmail services, Hotmail.com was founded by Sabeer Bhatia and Jack Smith in Mountain View, California with their only

competitor at the time being RocketMail (later Yahoo! Mail). It was commercially launched on July 4, 1996 to coincide with the American Independence Day. This was done to emphasise the freedom from ISP based email that Hotmail provided, along with the ability to access the mail from anywhere in the world. Microsoft acquired Hotmail for \$400 million in December 1997 and later rebranded it as MSN Hotmail.



THE FIFTH STATE OF MATTER

If you aren't shocked that a fifth state exists, you probably know that it is called the Bose-Einstein Condensate. Named after Satyendra Nath Bose and Albert Einstein, this state comprises of a dilute gas of bosons (also named after S.N.Bose) cooled down to absolute zero where they stay at the lowest quantum state. This state has significant applications in quantum computing where bringing down the particles to a single quantum state is highly important, and also in superconductivity.

S.N.Bose had sent a paper to Albert Einstein about light quanta where he derived Planck's quantum radiation law without referring to classical physics. This impressed Einstein who further expanded the ideas to two other papers. This combined effort resulted in the conceptualisation of a Bose gas. In 1995 the first condensate was created, shortly after which another independent



Bust of S.N.Bose at Birla Industrial & Technological Museum.



The exhibit about Mr. Vinod Dham at the Smithsonian